From the 2017 Administration

AP® CollegeBoard

Physics 2: Algebra-Based

Practice Exam

NOTE: This is a modified version of the 2017 AP Physics 2: Algebra-Based Exam.

AP® PHYSICS 2 TABLE OF INFORMATION

CONSTANTS AND CONVERSION FACTORS	
Proton mass, $m_p = 1.67 \times 10^{-27}$ kg	Electron charge magnitude, $e = 1.60 \times 10^{-19}$ C
Neutron mass, $m_n = 1.67 \times 10^{-27}$ kg	1 electron volt, $1 \text{ eV} = 1.60 \times 10^{-19}$ J
Electron mass, $m_e = 9.11 \times 10^{-31}$ kg	Speed of light, $c = 3.00 \times 10^8$ m/s
Avogadro's number, $N_0 = 6.02 \times 10^{23} \text{ mol}^{-1}$	Universal gravitational constant, $G = 6.67 \times 10^{-11} \text{ m}^3/\text{kg}\cdot\text{s}^2$
Universal gas constant, $R = 8.31$ J/(mol•K)	Acceleration due to gravity at Earth's surface, $g = 9.8 \text{ m/s}^2$
Boltzmann's constant, $k_B = 1.38 \times 10^{-23}$ J/K	

1 unified atomic mass unit,	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg} = 931 \text{ MeV}/c^2$
Planck's constant,	$h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s} = 4.14 \times 10^{-15} \text{ eV}\cdot\text{s}$
	$hc = 1.99 \times 10^{-25} \text{ J}\cdot\text{m} = 1.24 \times 10^3 \text{ eV}\cdot\text{nm}$
Vacuum permittivity,	$\varepsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2$
Coulomb's law constant,	$k = 1/4\pi\varepsilon_0 = 9.0 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$
Vacuum permeability,	$\mu_0 = 4\pi \times 10^{-7}$ (T•m)/A
Magnetic constant,	$k' = \mu_0/4\pi = 1 \times 10^{-7}$ (T•m)/A
1 atmosphere pressure,	$1 \text{ atm} = 1.0 \times 10^5 \text{ N/m}^2 = 1.0 \times 10^5$ Pa

UNIT SYMBOLS								
	meter,	m	mole,	mol	watt,	W	farad,	F
	kilogram,	kg	hertz,	Hz	coulomb,	C	tesla,	T
	second,	s	newton,	N	volt,	V	degree Celsius,	°C
	ampere,	A	pascal,	Pa	ohm,	Ω	electron volt,	eV
	kelvin,	K	joule,	J	henry,	H		

PREFIXES		
Factor	Prefix	Symbol
10^{12}	tera	T
10^{9}	giga	G
10^{6}	mega	M
10^{3}	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p

VALUES OF TRIGONOMETRIC FUNCTIONS FOR COMMON ANGLES

θ	$0°$	$30°$	$37°$	$45°$	$53°$	$60°$	$90°$
$\sin\theta$	0	$1/2$	$3/5$	$\sqrt{2}/2$	$4/5$	$\sqrt{3}/2$	1
$\cos\theta$	1	$\sqrt{3}/2$	$4/5$	$\sqrt{2}/2$	$3/5$	$1/2$	0
$\tan\theta$	0	$\sqrt{3}/3$	$3/4$	1	$4/3$	$\sqrt{3}$	∞

The following conventions are used in this exam.

I. The frame of reference of any problem is assumed to be inertial unless otherwise stated.
II. In all situations, positive work is defined as work done on a system.
III. The direction of current is conventional current: the direction in which positive charge would drift.
IV. Assume all batteries and meters are ideal unless otherwise stated.
V. Assume edge effects for the electric field of a parallel plate capacitor unless otherwise stated.
VI. For any isolated electrically charged object, the electric potential is defined as zero at infinite distance from the charged object

AP® PHYSICS 2 EQUATIONS

MECHANICS

$$v_x = v_{x0} + a_x t$$

$$x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$$

$$v_x^2 = v_{x0}^2 + 2a_x (x - x_0)$$

$$\vec{a} = \frac{\sum \vec{F}}{m} = \frac{\vec{F}_{net}}{m}$$

$$\left|\vec{F}_f\right| \le \mu \left|\vec{F}_n\right|$$

$$a_c = \frac{v^2}{r}$$

$$\vec{p} = m\vec{v}$$

$$\Delta\vec{p} = \vec{F}\,\Delta t$$

$$K = \frac{1}{2} m v^2$$

$$\Delta E = W = F_{\parallel} d = F d \cos\theta$$

$$P = \frac{\Delta E}{\Delta t}$$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$$

$$\omega = \omega_0 + \alpha t$$

$$x = A\cos(\omega t) = A\cos(2\pi f t)$$

$$x_{cm} = \frac{\sum m_i x_i}{\sum m_i}$$

$$\vec{\alpha} = \frac{\sum \vec{\tau}}{I} = \frac{\vec{\tau}_{net}}{I}$$

$$\tau = r_{\perp} F = r F \sin\theta$$

$$L = I\omega$$

$$\Delta L = \tau\,\Delta t$$

$$K = \frac{1}{2} I \omega^2$$

$$\left|\vec{F}_s\right| = k\left|\vec{x}\right|$$

a = acceleration
A = amplitude
d = distance
E = energy
F = force
f = frequency
I = rotational inertia
K = kinetic energy
k = spring constant
L = angular momentum
ℓ = length
m = mass
P = power
p = momentum
r = radius or separation
T = period
t = time
U = potential energy
v = speed
W = work done on a system
x = position
y = height
α = angular acceleration
μ = coefficient of friction
θ = angle
τ = torque
ω = angular speed

$$U_s = \frac{1}{2} k x^2$$

$$\Delta U_g = m g \,\Delta y$$

$$T = \frac{2\pi}{\omega} = \frac{1}{f}$$

$$T_s = 2\pi\sqrt{\frac{m}{k}}$$

$$T_p = 2\pi\sqrt{\frac{\ell}{g}}$$

$$\left|\vec{F}_g\right| = G\frac{m_1 m_2}{r^2}$$

$$\vec{g} = \frac{\vec{F}_g}{m}$$

$$U_G = -\frac{G m_1 m_2}{r}$$

ELECTRICITY AND MAGNETISM

$$\left|\vec{F}_E\right| = \frac{1}{4\pi\varepsilon_0}\frac{\left|q_1 q_2\right|}{r^2}$$

$$\vec{E} = \frac{\vec{F}_E}{q}$$

$$\left|\vec{E}\right| = \frac{1}{4\pi\varepsilon_0}\frac{|q|}{r^2}$$

$$\Delta U_E = q\Delta V$$

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$$

$$\left|\vec{E}\right| = \left|\frac{\Delta V}{\Delta r}\right|$$

$$\Delta V = \frac{Q}{C}$$

$$C = \kappa\varepsilon_0\frac{A}{d}$$

$$E = \frac{Q}{\varepsilon_0 A}$$

$$U_C = \frac{1}{2} Q\Delta V = \frac{1}{2} C(\Delta V)^2$$

$$I = \frac{\Delta Q}{\Delta t}$$

$$R = \frac{\rho\ell}{A}$$

$$P = I\,\Delta V$$

$$I = \frac{\Delta V}{R}$$

$$R_s = \sum_i R_i$$

$$\frac{1}{R_p} = \sum_i \frac{1}{R_i}$$

$$C_p = \sum_i C_i$$

$$\frac{1}{C_s} = \sum_i \frac{1}{C_i}$$

$$B = \frac{\mu_0}{2\pi}\frac{I}{r}$$

A = area
B = magnetic field
C = capacitance
d = distance
E = electric field
$\mathcal{E}$ = emf
F = force
I = current
ℓ = length
P = power
Q = charge
q = point charge
R = resistance
r = separation
t = time
U = potential (stored) energy
V = electric potential
v = speed
κ = dielectric constant
ρ = resistivity
θ = angle
Φ = flux

$$\vec{F}_M = q\vec{v} \times \vec{B}$$

$$\left|\vec{F}_M\right| = \left|q\vec{v}\right|\left|\sin\theta\right|\left|\vec{B}\right|$$

$$\vec{F}_M = I\vec{\ell} \times \vec{B}$$

$$\left|\vec{F}_M\right| = \left|I\vec{\ell}\right|\left|\sin\theta\right|\left|\vec{B}\right|$$

$$\Phi_B = \vec{B}\bullet\vec{A}$$

$$\Phi_B = \left|\vec{B}\right|\cos\theta\left|\vec{A}\right|$$

$$\mathcal{E} = -\frac{\Delta\Phi_B}{\Delta t}$$

$$\mathcal{E} = B\ell v$$

AP® PHYSICS 2 EQUATIONS

FLUID MECHANICS AND THERMAL PHYSICS

$$\rho = \frac{m}{V}$$

$$P = \frac{F}{A}$$

$$P = P_0 + \rho g h$$

$$F_b = \rho V g$$

$$A_1 v_1 = A_2 v_2$$

$$P_1 + \rho g y_1 + \frac{1}{2}\rho v_1^2 = P_2 + \rho g y_2 + \frac{1}{2}\rho v_2^2$$

$$\frac{Q}{\Delta t} = \frac{kA\,\Delta T}{L}$$

$$PV = nRT = Nk_B T$$

$$K = \frac{3}{2}k_B T$$

$$W = -P\,\Delta V$$

$$\Delta U = Q + W$$

A = area
F = force
h = depth
k = thermal conductivity
K = kinetic energy
L = thickness
m = mass
n = number of moles
N = number of molecules
P = pressure
Q = energy transferred to a system by heating
T = temperature
t = time
U = internal energy
V = volume
v = speed
W = work done on a system
y = height
ρ = density

WAVES AND OPTICS

$$\lambda = \frac{v}{f}$$

$$n = \frac{c}{v}$$

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

$$\frac{1}{s_i} + \frac{1}{s_o} = \frac{1}{f}$$

$$|M| = \left|\frac{h_i}{h_o}\right| = \left|\frac{s_i}{s_o}\right|$$

$$\Delta L = m\lambda$$

$$d\sin\theta = m\lambda$$

d = separation
f = frequency or focal length
h = height
L = distance
M = magnification
m = an integer
n = index of refraction
s = distance
v = speed
λ = wavelength
θ = angle

MODERN PHYSICS

$$E = hf$$

$$K_{\max} = hf - \phi$$

$$\lambda = \frac{h}{p}$$

$$E = mc^2$$

E = energy
f = frequency
K = kinetic energy
m = mass
p = momentum
λ = wavelength
ϕ = work function

GEOMETRY AND TRIGONOMETRY

Rectangle
$A = bh$

Triangle
$A = \frac{1}{2}bh$

Circle
$A = \pi r^2$
$C = 2\pi r$

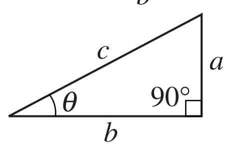
Rectangular solid
$V = \ell w h$

Cylinder
$V = \pi r^2 \ell$
$S = 2\pi r \ell + 2\pi r^2$

Sphere
$V = \frac{4}{3}\pi r^3$
$S = 4\pi r^2$

A = area
C = circumference
V = volume
S = surface area
b = base
h = height
ℓ = length
w = width
r = radius

Right triangle
$c^2 = a^2 + b^2$
$\sin\theta = \frac{a}{c}$
$\cos\theta = \frac{b}{c}$
$\tan\theta = \frac{a}{b}$



PHYSICS 2

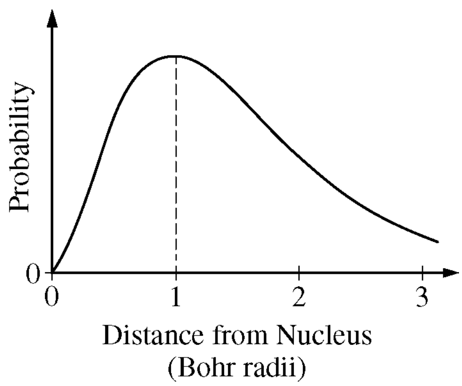
Section I

40 Questions

Time—90 minutes

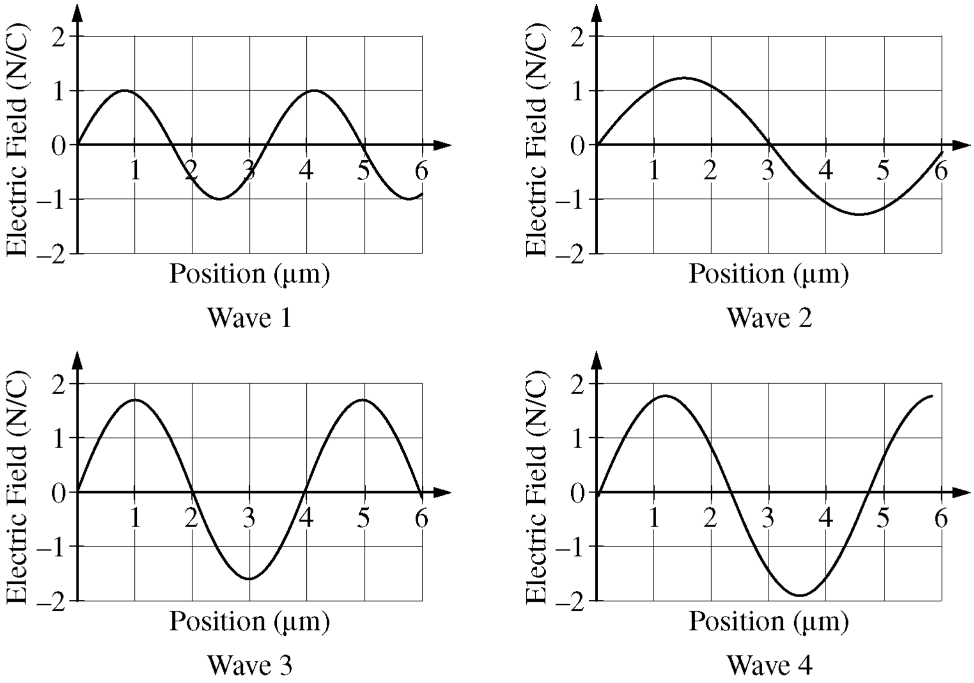
Note: To simplify calculations, you may use $g = 10 \text{ m/s}^2$ in all problems.

Directions: Each of the questions or incomplete statements below is followed by four suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.



1. The graph above describes the location of an electron in a hydrogen atom that is in the ground state. What conclusion can be drawn from the graph?

   (A) The distance of the electron from the nucleus is exactly equal to one Bohr radius.
   (B) The electron will probably slide down either the left slope or the right slope, depending on its initial location.
   (C) The electron must always be located within three Bohr radii of the nucleus.
   (D) The greatest probability of locating the electron is at a distance of one Bohr radius from the nucleus.



2. The graphs above show the electric field as a function of position at some instant of time for four different electromagnetic plane waves. Which of the following correctly ranks the frequency f of the waves?

(A) $f_1 > f_2 > f_3 > f_4$

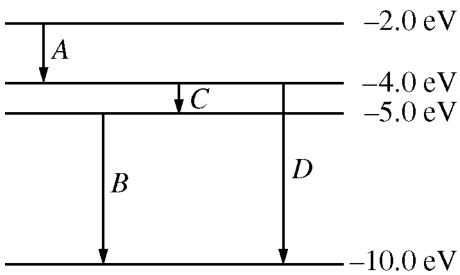
(B) $f_1 > f_3 > f_4 > f_2$

(C) $f_2 > f_4 > f_3 > f_1$

(D) $f_4 > f_3 > f_2 > f_1$

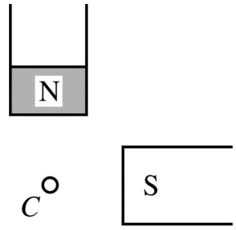
3. When two bowling balls are held with their centers 30 centimeters apart, they exert a gravitational force of magnitude F on each other. For two free protons to exert a net force of magnitude F on each other, the protons must be separated by a distance of approximately 1×10^{-10} m. Under these conditions, will the protons attract or repel each other, and why?

(A) They will attract each other, because they both have positive charge.
(B) They will attract each other, because the gravitational force is always attractive.
(C) They will repel each other, because the repulsive electrical force between the protons is much stronger than the attractive gravitational force.
(D) They will repel each other, because the protons only exert a repulsive electrical force on each other and do not exert an attractive gravitational force.

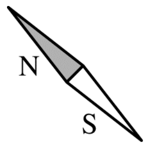


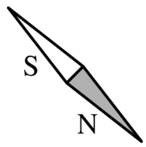
4. The energy level diagram for a hypothetical atom is shown in the figure above. Which of the labeled transitions in the diagram would produce a photon with the shortest wavelength?

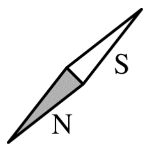
(A) A
(B) B
(C) C
(D) D

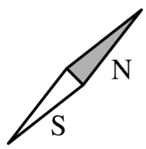


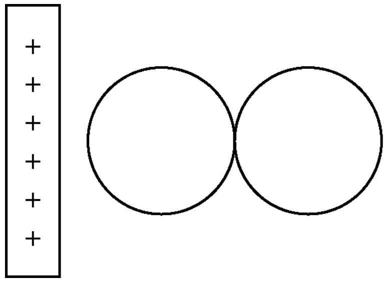
5. The figure above shows a small compass C that is equidistant from the poles of two identical bar magnets. Which of the following best represents the direction that the needle of the compass will point?

(A) 

(B) 

(C) 

(D) 



6. Two uncharged conducting spheres on insulating stands are in contact when a positively charged rod is brought near but does not touch them, as shown in the figure above. The right sphere is then moved far to the right, and finally the rod is removed. Which of the following best shows the final distribution of excess charges, if any, on the spheres?

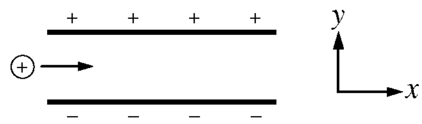
Left Sphere — Right Sphere

(A) 

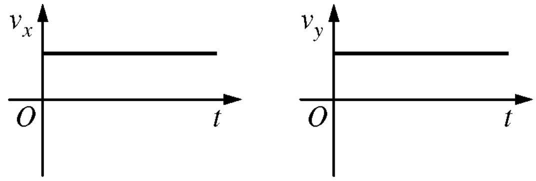
(B) 

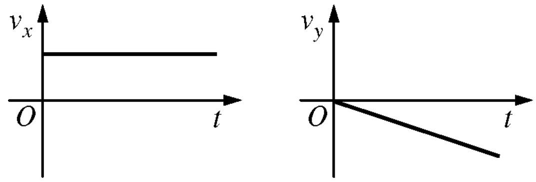
(C) 

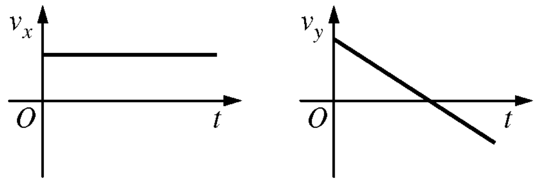
(D) 

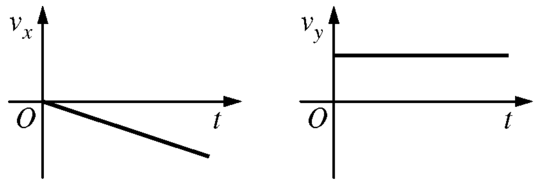


7. A positively charged particle is moving horizontally when it enters the uniform electric field between two parallel charged plates, as shown in the figure above. Which of the following correctly shows the x- and y-components of the velocity v as a function of time t ?

(A) 

(B) 

(C) 

(D) 

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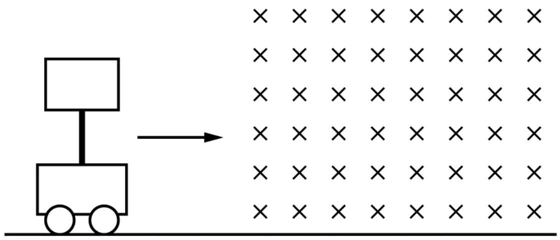
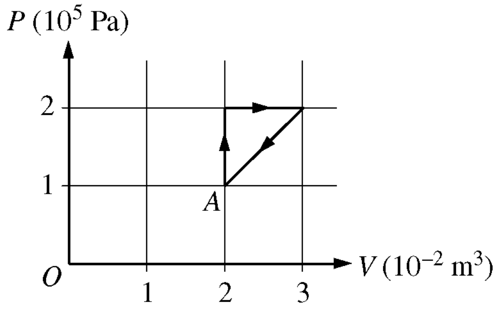
8. A simple circuit has a 24.0 V battery and resistors of resistance $3.0\ \Omega$, $6.0\ \Omega$, and R all connected in parallel. If the current through the battery is 15.0 A, what is the value of R ?

(A) $0.9\ \Omega$
(B) $1.1\ \Omega$
(C) $7.4\ \Omega$
(D) $8.0\ \Omega$



9. A rectangular loop of copper wire is attached to a cart by an insulating rod. The cart is moving at constant speed when it enters a region containing a uniform magnetic field that is perpendicular to the plane of the loop and directed into the page, as shown above. Frictional losses are negligible. Which of the following correctly describes the speed of the cart as it moves into, through, and out of the field?

(A) The speed remains constant.
(B) The speed continually decreases.
(C) The speed decreases as the cart enters the field and increases as it leaves the field but is constant while it is completely inside the field.
(D) The speed decreases as the cart enters and leaves the field but is constant while it is completely inside the field.



10. A sample of ideal gas that is initially in state A shown on the graph above is taken through the cycle shown. What is the net work done on the gas during one cycle?

(A) -500 J
(B) -1000 J
(C) -2000 J
(D) -3500 J

11. A small, well-insulated building is kept cool on a hot day. A person in the building holds a hand close to, but not touching, a solid metal door to the outside and feels warmth coming from the metal door. Which statement best describes how most of the energy is transferred from the outside to the person's hand?

(A) Thermal energy from the outside is convected through the metal door. Thermal energy from the door is conducted by the air and is detected by the person's hand.
(B) Thermal energy from the outside is convected through the metal door. The door radiates energy that is detected by the person's hand.
(C) Thermal energy from the outside is conducted through the metal door. The door radiates energy that is detected by the person's hand.
(D) Thermal energy from the outside is radiated through the metal door. The door radiates energy that is detected by the person's hand.

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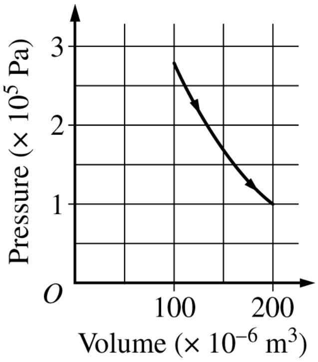
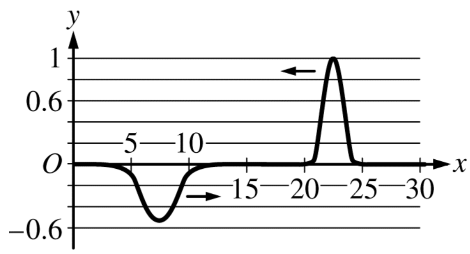


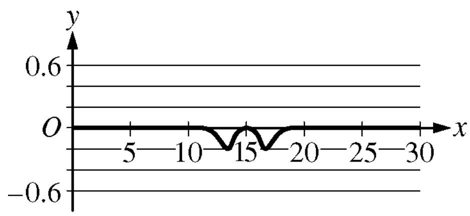
12. The graph above of pressure as a function of volume illustrates a thermodynamic process for an ideal gas in an engine. Which of the following best describes the change in the internal energy of the gas and the work done during the process?

    (A) The internal energy increases, and positive work is done by the gas.
    (B) The internal energy decreases, and positive work is done on the gas.
    (C) The internal energy decreases, and positive work is done by the gas.
    (D) The internal energy increases, and positive work is done on the gas.

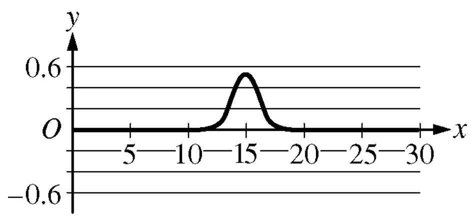


13. Two wave pulses approach each other, as shown in the figure above. The pulse traveling to the right is half the height and twice the width of the pulse traveling to the left. Which of the following best represents the resulting shape when the peaks of the two pulses are at the same location?

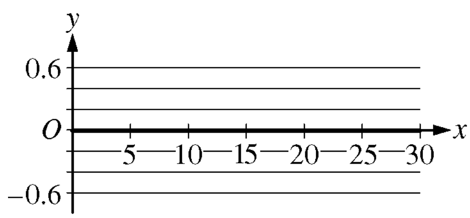
(A)



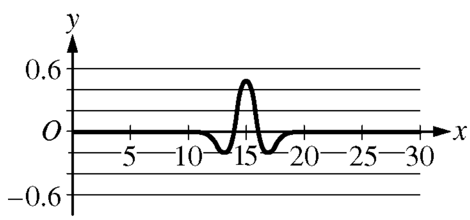
(B)



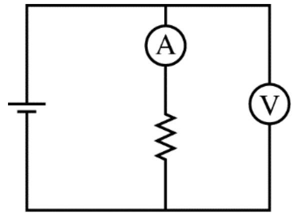
(C)



(D)

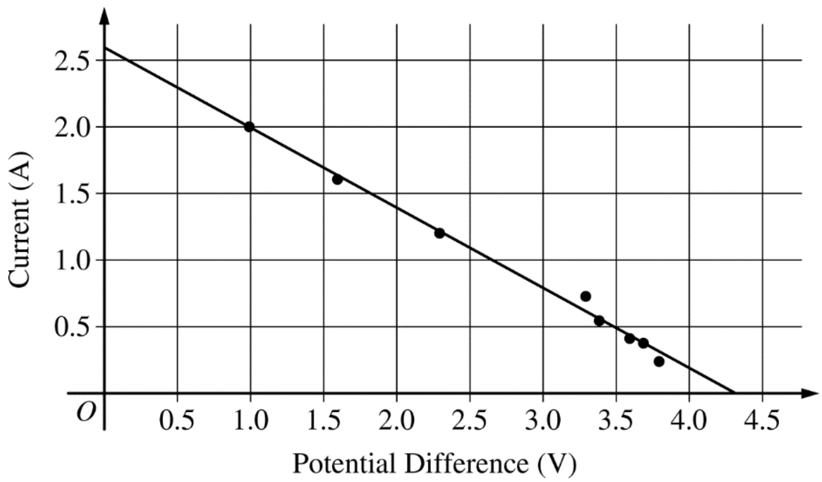


Questions 14-15 refer to the following material.



A student performs an experiment using the circuit shown above, with the goal of determining the characteristics of the battery, which has nonnegligible internal resistance. The student has many resistors with resistances ranging from $0.5\ \Omega$ to $15\ \Omega$. The student puts different resistors in the circuit and records the current and the potential difference measured by the ammeter A and voltmeter V. The student's data table and graph of current as a function of potential difference are shown below.

Resistance (Ω)	0.5	1.0	2.0	4.0	6.0	8.0	10	15
Potential Difference (V)	1.0	1.6	2.3	3.3	3.4	3.6	3.7	3.8
Current (A)	2.0	1.6	1.2	0.72	0.55	0.41	0.35	0.23



14. Which of the following is most nearly the battery's internal resistance?

    (A) 0.4 ohm
    (B) 0.6 ohm
    (C) 1.6 ohm
    (D) 2.6 ohm

15. The student now replaces the battery with a different one that has the same emf but a significantly smaller internal resistance. The student runs the experiment again and graphs the new data. Which of the following correctly describes how the new graph compares to the plot shown above?

    (A) The new graph will be the same as the original graph.
    (B) The vertical intercept will have a greater value, but the slope will be the same.
    (C) The vertical intercept will be the same, but the magnitude of the slope will be greater.
    (D) Both the vertical intercept and the magnitude of the slope will be greater.

16. A sample of gas has a temperature of 200 K. If the speed of every gas molecule in the sample is doubled, what is the new temperature of the gas?

(A) 800 K
(B) 400 K
(C) 200 K
(D) 100 K

Questions 17-19 refer to the following material.

An atom of polonium (Po-216) is moving slowly enough that it can be considered to be at rest. The Po-216 undergoes alpha decay and becomes lead (Pb-212), via the reaction ${}^{216}_{84}\text{Po} \rightarrow {}^{212}_{82}\text{Pb} + {}^{4}_{2}\alpha$. After the decay, the lead atom is moving to the left with speed v_{Pb}, and the alpha particle is moving to the right with speed v_{α}. The masses of the three isotopes involved in the decay are given below.

$$M_{\text{Po-216}} = 216.001915\ \text{u}$$

$$M_{\alpha} = 4.002603\ \text{u}$$

$$M_{\text{Pb-212}} = 211.991898\ \text{u}$$

17. How do the momentum and kinetic energy of the polonium atom compare with the total momentum and kinetic energy of the decay products?

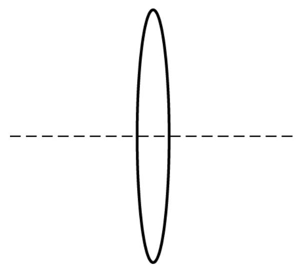
	Polonium Momentum	Polonium Kinetic Energy
(A)	Different	Different
(B)	Different	The same
(C)	The same	Different
(D)	The same	The same

18. How much energy is released in the decay?

(A) None
(B) 0.00741 MeV
(C) 6.90 MeV
(D) 9.31 MeV

19. The half-life of Po-216 is $1/7$ s. What is the probability that any particular Po-216 atom will decay within one second?

(A) 0 %
(B) 7 %
(C) 99 %
(D) 100 %



20. The figure above shows a lens with a focal length of magnitude 30 cm when in air. An object is located 200 cm to the left of the lens. What is the approximate location of the object's image?

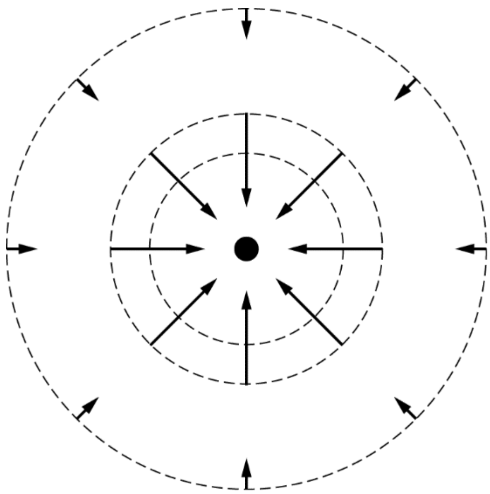
(A) 26 cm to the left of the lens
(B) 26 cm to the right of the lens
(C) 35 cm to the left of the lens
(D) 35 cm to the right of the lens

Questions 21-23 refer to the following material.

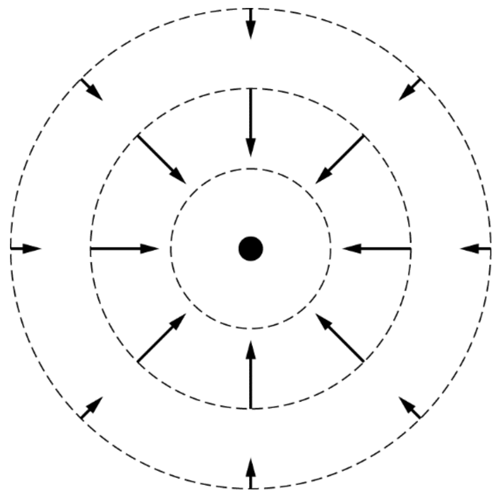
A very small, isolated sphere with charge $+Q$ exists in an empty region of space. A second very small sphere is moved from far away to a short distance from the first sphere.

21. In the figures below, the central dot represents the first sphere. The dashed circles are isolines with the same potential difference between adjacent circles. Which of the figures best represents the isolines and electric field vectors around the first sphere while the second sphere is still far away?

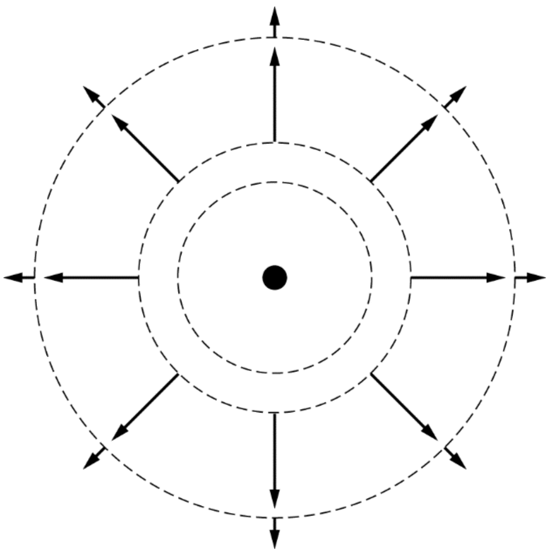
(A)



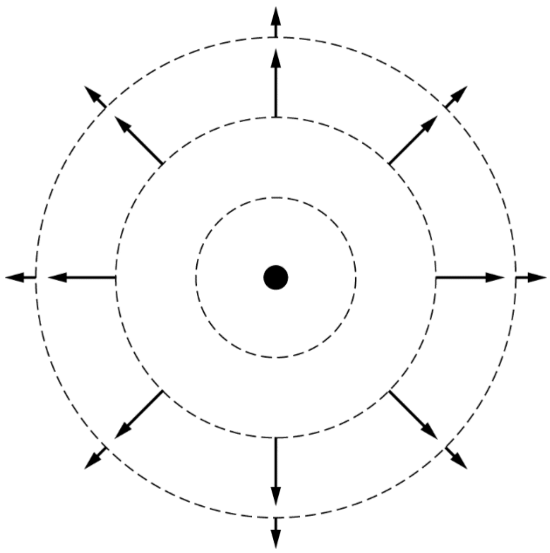
(B)



(C)



(D)



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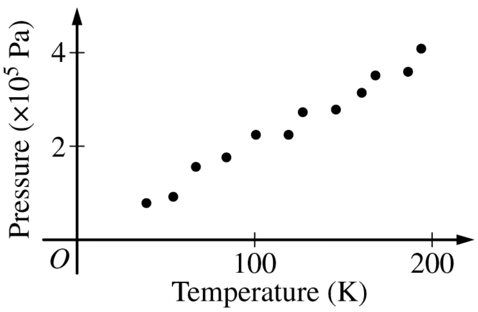
22. The second sphere has a charge of $+2.0 \times 10^{-9}$ C. As it is moved closer to the first sphere at a constant speed, the second sphere passes through the circular equipotential lines due to the first sphere. Two of these lines are separated by a distance of 0.020 m and have potentials of 100 V and 150 V. What is the magnitude of the average force needed to move the second sphere between the two equipotential lines?

(A) 1.5×10^{-5} N
(B) 5.0×10^{-6} N
(C) 1.0×10^{-7} N
(D) 2.0×10^{-9} N

23. As the second sphere is moved at a constant velocity closer to the first sphere, data is taken for the force F exerted on the second sphere by the first sphere as a function of the separation d of the spheres. Which of the following should be graphed to yield a straight line from the data?

(A) F as a function of d
(B) F as a function of d^2
(C) F as a function of $1/d$
(D) F as a function of $1/d^2$



24. A sample of gas is in a sealed container whose volume is held fixed. The pressure of the gas is measured as its absolute temperature is increased. A graph of pressure measurements as a function of temperature is shown above. According to the ideal gas law, which of the following is expected if the sealed container's volume is increased to three times the original value and the experiment is repeated?

(A) The graph will show a linear relationship that still extrapolates to the origin, but with a smaller slope.
(B) The graph will show a linear relationship that still extrapolates to the origin, but with a larger slope.
(C) The graph will show a linear relationship with the same slope, but it will extrapolate to a point on the horizontal axis to the right of the origin.
(D) The graph will show a linear relationship with the same slope, but it will extrapolate to a point on the vertical axis above the origin.

25. Which of the following describes a difference in the behavior of an electrically conducting sphere and that of an insulating sphere?

(A) A conducting sphere can be charged by friction, but an insulating sphere cannot.
(B) An uncharged object can be charged by touching it to a charged conducting sphere, but not by touching it to a charged insulating sphere.
(C) When a conducting sphere is brought near a positively charged object, some of the sphere's electrons move closer to that object. No polarization occurs in the atoms of an insulating sphere.
(D) Excess charge placed on a conducting sphere becomes distributed over the entire surface of the sphere. Excess charge placed on an insulating sphere can remain where it is placed.

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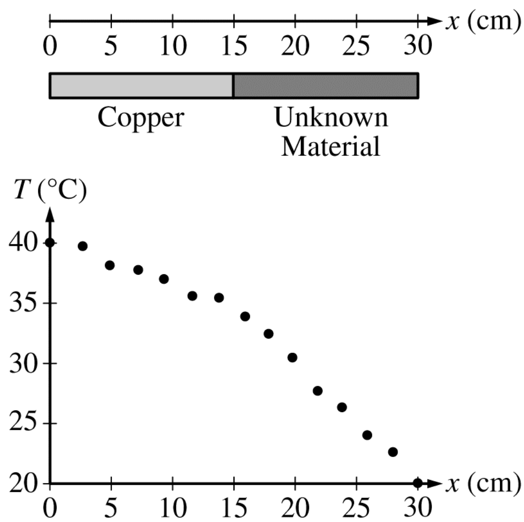


26. A student designs an experiment intended to determine the thermal conductivity of an unknown material. Two 15 cm long bars with the same cross-sectional area are joined together, as shown in the figure above. The bar on the left is made of copper, and the bar on the right is made of the unknown material. The left and right ends of the two-bar combination are maintained at $40°C$ and $20°C$, respectively, and the bars reach equilibrium. The student then measures the temperature T along the bars and produces the data shown in the graph, where $x = 0$ is the left end and 30 cm is the right end. The experimental uncertainty in the data is negligible. Which of the following can the student conclude about k_c, the thermal conductivity of copper, compared with k_u, the thermal conductivity of the unknown material?

(A) $k_c < k_u$
(B) $k_c = k_u$
(C) $k_c > k_u$
(D) The relationship cannot be determined from the data given.

27. A sealed container with a lid of area $0.004\ m^2$ is filled with an ideal gas. The container and gas are allowed to reach thermal equilibrium with the surrounding air. If a 2000 N block is needed to keep the lid from being pushed off the container, what is the absolute pressure inside the container (the pressure compared to vacuum) ?

(A) 1×10^5 Pa
(B) 2×10^5 Pa
(C) 4×10^5 Pa
(D) 6×10^5 Pa

28. Water initially moving through a small horizontal cylindrical pipe enters a new section of pipe with twice the radius of the initial pipe. What is the ratio of the speed of the water in the new section of pipe to the initial speed of the water?

(A) 2/1
(B) 1/1
(C) 1/2
(D) 1/4

29. A block floating in water is supported by the buoyant force exerted on the block by the water. The buoyant force is created on the atomic scale primarily by which of the following electrostatic forces?

(A) A force of attraction between neutral atoms of the block and neutral atoms of the water
(B) A force of attraction between charged atoms of the block and charged atoms of the water
(C) A force of repulsion between nuclei in the neutral atoms of the block and nuclei in the neutral atoms of the water
(D) A force of repulsion between electrons in the neutral atoms of the block and electrons in the neutral atoms of the water

30. A supernova explosion occurs at a distance of 2×10^{20} m from Earth. The supernova simultaneously emits radiation at both visible and x-ray wavelengths. Which of the following statements best describes the time it takes for the two types of radiation from the supernova to reach Earth?

(A) The two types of radiation reach Earth at the same time, within a few seconds of the supernova explosion.
(B) The two types of radiation reach Earth at the same time, many years after the supernova explosion.
(C) The two types of radiation reach Earth many years after the supernova explosion, but the visible radiation arrives first due to its shorter wavelength.
(D) The two types of radiation reach Earth many years after the supernova explosion, but the x-ray radiation arrives first due to its shorter wavelength.

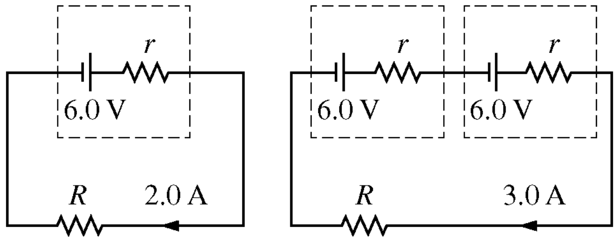
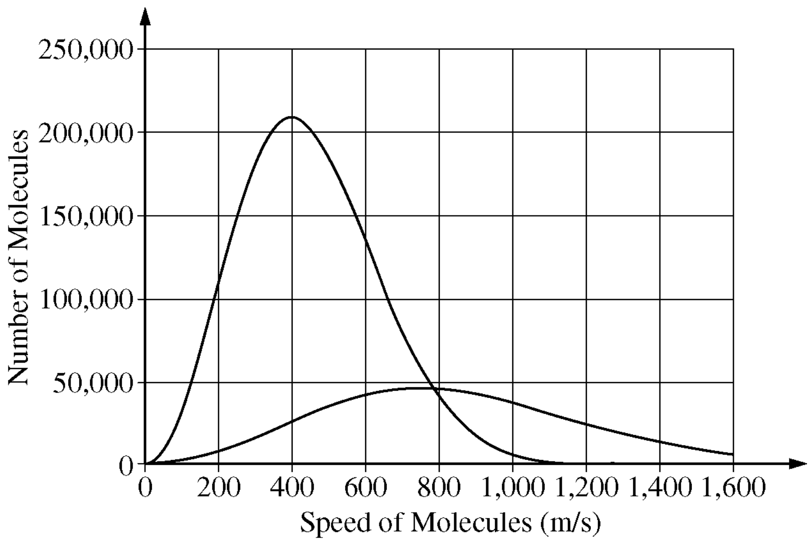


Figure 1 Figure 2

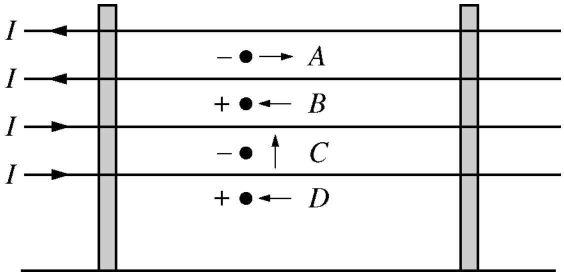
31. A battery with emf 6.0 V and an internal resistance r is connected to a resistor of resistance R, as shown in Figure 1 above. The current in the circuit is 2.0 A. When an identical battery is added to the circuit in series, as shown in Figure 2, the current in the circuit is 3.0 A. Which of the following statements about the resistances in the circuit is true?

(A) r is very small compared to R.
(B) r is very large compared to R.
(C) r is equal to R.
(D) r and R are comparable in magnitude, but they are not equal.



32. Two samples of the same type of gas molecules are placed in different closed containers and are thermally isolated from each other and the environment. The figure above shows the distribution of speeds for the molecules in each of the samples. The two samples are now mixed together and allowed to reach thermal equilibrium. Which of the following is true of the peak of the distribution for the mixed sample?

(A) It is closer to $400\ \text{m/s}$ than it is to $800\ \text{m/s}$.
(B) It is closer to $800\ \text{m/s}$ than it is to $400\ \text{m/s}$.
(C) It is midway between the peaks of the distributions shown.
(D) There is not just one peak but two, at the same locations as the peaks of the distributions shown.



33. Four horizontal wires are arranged on vertical wooden poles, as shown in the figure above. The wires are equally spaced and have equal currents I in the directions indicated in the figure. The dots represent four charged dust particles moving in the plane of the wires. The sign of the charge on each particle and its direction of motion at a particular instant are shown. Which of the dust particles has a magnetic force exerted on it in the downward direction at this instant?

(A) A
(B) B
(C) C
(D) D

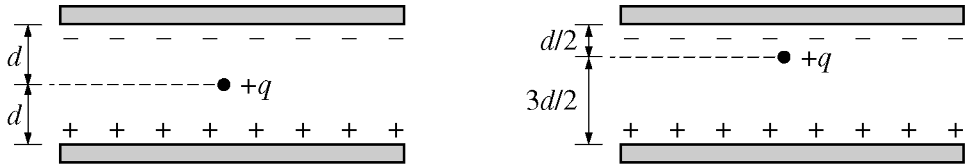
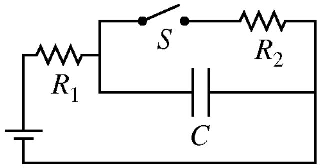


Figure 1 Figure 2

34. A parallel-plate capacitor is charged so that an electric field E is created between the plates. A small oil drop with charge $+q$ is placed halfway between the plates, as shown in Figure 1 above. The force exerted on the oil drop by the electric field is determined to have a magnitude of F. The oil drop is now moved half the distance toward the negatively charged plate, as shown in Figure 2 above. What is the magnitude of the force exerted by the electric field on the oil drop at the position shown in Figure 2 ?

(A) $F/2$
(B) F
(C) $2F$
(D) $4F$



35. In the circuit represented above, resistors R_1 and R_2, capacitor C, and open switch S are connected to a battery. The circuit reaches equilibrium. The switch is then closed, and the circuit is allowed to come to a new equilibrium. Which of the following is a true statement about the charge q_f stored on the capacitor after the switch is closed compared with the charge q_i stored on the capacitor before the switch is closed?

(A) $q_f > q_i$
(B) $q_f = q_i$
(C) $q_f < q_i$
(D) The relationship between the charges cannot be determined unless the resistances of the resistors are known.

36. Louis de Broglie proposed that all forms of matter have both wave properties and particle properties. Which of the following explains why diffraction effects are observable only for small-scale objects?

(A) Wavelengths of large-scale objects are much smaller than any aperture through which the objects could pass.
(B) Wavelengths of large-scale objects are much larger than any aperture through which the objects could pass.
(C) Large-scale objects have too much energy to allow observation of their wave properties.
(D) Large-scale objects move too slowly to allow observation of their wave properties.

Directions: For each of the questions or incomplete statements below, two of the suggested answers will be correct. For each of these questions, you must select both correct choices to earn credit. No partial credit will be earned if only one correct choice is selected. Select the two that are best in each case and then fill in the corresponding circles that begin with number 131 on page 3 of the answer sheet.

131. A student is given a sample of gas that is at room temperature and is sealed in a container by a movable piston. Which of the following adjustments would allow the student to increase the pressure of the gas? Select two answers.

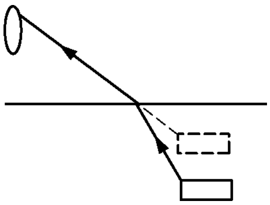
(A) Put the container in a tank of room-temperature water and pull the piston outward.
(B) Put the container in a tank of room-temperature water and push the piston inward.
(C) Put the container in a tank of hot water and hold the piston in place.
(D) Put the container in a tank of cold water and hold the piston in place.

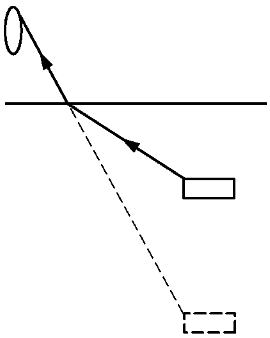
132. A student is given two different convex spherical mirrors and asked to determine which of the mirrors has the shorter focal length. Answering which of the following questions would allow the student to make this determination? Select two answers.

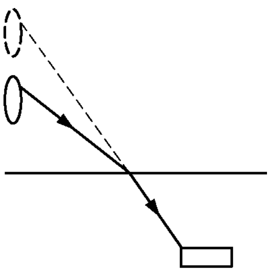
(A) Which mirror has a larger magnification for a given object distance?
(B) Which mirror has the greater change in magnification when submerged in water?
(C) Which mirror produces an upright image?
(D) Which mirror has a smaller radius of curvature?

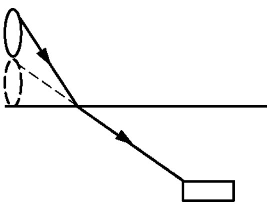
GO ON TO THE NEXT PAGE.

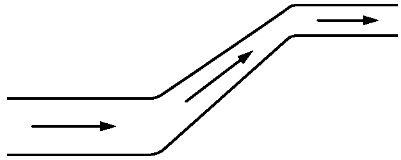
133. On a calm day, student 1 looks down into a pool and sees student 2 swimming underwater. At the same time, student 2 looks up from under the water and sees student 1. In the following diagrams, the oval represents student 1 and the rectangle represents student 2. Diagrams that correctly illustrate the apparent positions seen by student 1 and student 2 include which of the following? Select two answers.

(A)


(B)


(C)


(D)




134. An ideal fluid is moving from left to right through a pipe, as shown in the figure above. The diameter of the pipe decreases and the height increases from left to right. True statements about the pressure of the fluid as it moves through the pipe include which of the following? Select two answers.

(A) The pressure tends to decrease due to the increase in the fluid's speed as the pipe gets narrower.
(B) The pressure tends to increase due to the decrease in the fluid's speed as the pipe gets narrower.
(C) The pressure tends to decrease due to the increase in the potential energy of the fluid-Earth system as the pipe gets higher.
(D) The pressure tends to increase due to the increase in the potential energy of the fluid-Earth system as the pipe gets higher.

GO ON TO THE NEXT PAGE.

PHYSICS 2
Section II
4 Questions
Time—90 minutes

Directions: Questions 1 and 4 are short free-response questions that require about 20 minutes each to answer and are worth 10 points each. Questions 2 and 3 are long free-response questions that require about 25 minutes each to answer and are worth 12 points each. Show your work for each part in the space provided after that part.

1. (10 points, suggested time 20 minutes)

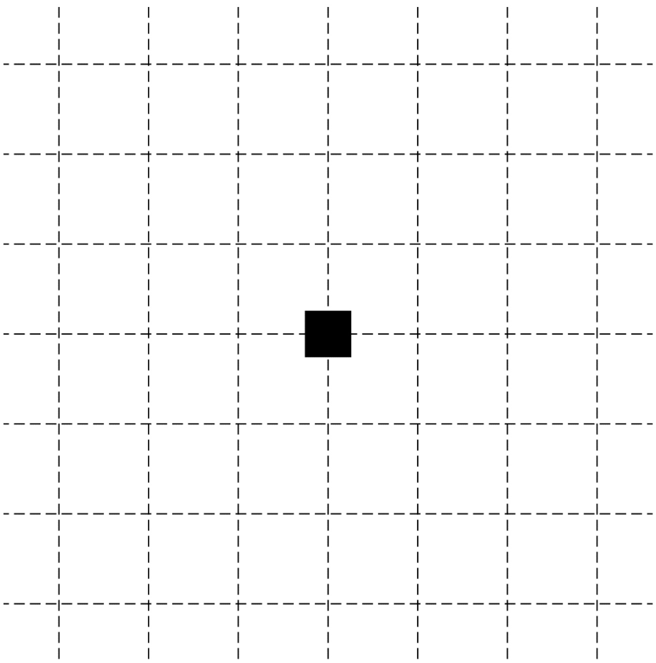
   A team of engineering students is testing their newly designed 200 kg raft in the pool where the diving team practices. The raft must hold a 730 kg steel cube with edges of length 45.0 cm without sinking. Assume the density of water in the pool is $1000\ \text{kg}/\text{m}^3$.

   (a) The students use a crane to gently place the cube on the raft but accidentally place it off center. The cube remains on the raft for a few moments and then the raft tilts, causing the cube to slide off and sink to the bottom of the pool. The raft remains floating in the pool. In a coherent paragraph-length response, indicate whether the water level in the pool when the cube is on the bottom of the pool is higher than, lower than, or the same as when the cube is on the raft, and explain your reasoning. For both cases, assume that there is no motion of the water.

(b) The bottom of the pool is 5 m below the surface of the water. The crane is now used to lift the cube out of the pool. The cube is lifted upward at a slow and constant speed, so all drag forces are negligible.

i. Predict how the force exerted by the crane on the cube when the bottom of the cube is at a depth of 2.0 m compares to the force exerted by the crane on the cube when the bottom of the cube is at a depth of 4.0 m. Explain your reasoning.

ii. On the black square below, which represents the cube, draw and label the forces (not components) that are exerted on the cube while the crane is lifting it and the bottom of the cube is 4 m below the surface of the water. Each force must be represented by a distinct arrow starting on, and pointing away from, the black square. The lengths of the arrows should approximately demonstrate the net force on the cube.



iii. Calculate the force exerted by the crane on the cube when the bottom of the cube is 4 m below the surface of the water.

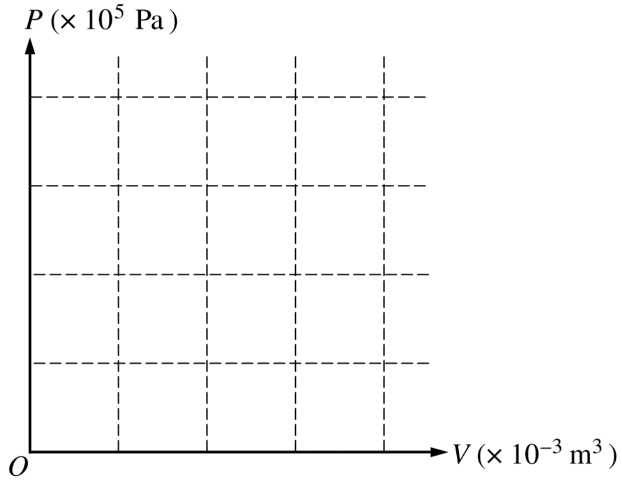
2. (12 points, suggested time 25 minutes)

A student gives the following description of a thermodynamic cycle through which a sample of an ideal monatomic gas is taken.

"Initially, the gas has a volume of $1 \times 10^{-3}\ \text{m}^3$ and a pressure of $1 \times 10^5\ \text{Pa}$. The gas is in a closed cylinder that has a movable piston on one end. Call the initial state of the gas state *A*. Process *AB* takes the gas at constant pressure from state *A* to state *B*, where *B* has three times the volume of state *A*. The gas is then compressed isothermally until it is back to its initial volume, reaching state *C*. Finally, a single process *CA* returns the gas to its initial state *A*."

(a)

i. On the axes below, plot states *A*, *B*, and *C*, and sketch a graph of pressure *P* as a function of volume *V* for cycle *ABCA*. Label states *A*, *B*, and *C*, and label the axes with numerical values.



ii. Let the temperature for state *A* be T_0. In terms of T_0, determine the temperature for state *B*.

iii. Compare the temperature at state *C* to the temperature at state *A*. Explain how your comparison is consistent with your graph above.

GO ON TO THE NEXT PAGE.

(b)

i. Briefly describe an experimental procedure that would take the gas from state C to state A.

ii. Relate the temperature of the gas to the microscopic processes involved in the energy transfer as the gas goes from state C to state A for the procedure you described in part (b)(i).

(c)

i. Use the student's verbal description or your graph in part (a)(i) to calculate a numerical value for the work done on the gas in process AB.

ii. The magnitude of the change in internal energy for process AB is $|\Delta U| = 300\text{ J}$. Calculate a numerical value for the energy added to or removed from the gas by heating in process AB.

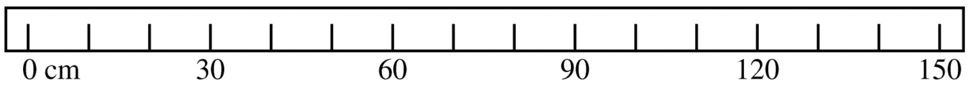
GO ON TO THE NEXT PAGE.

3. (12 points, suggested time 25 minutes)

Students are asked to perform an experiment to determine the focal length of a converging lens. They are given an optical bench that includes a lamp, a converging lens in a holder, and a screen. All three of these can be positioned anywhere along the optical bench. The students decide to form real images by placing the lamp at various specific distances from the lens and moving the screen until a focused image of the lamp filament is formed on the screen. They keep the lens at the 60 cm mark on the optical bench. The table below shows their data for the object and image distances.

Trial	1	2	3	4	5
Object Distance s_o (cm)	55	45	35	30	25
Image Distance s_i (cm)	21.3	23.8	28.5	32.5	43

(a) On the figure below, representing the optical bench, draw and label the screen, lens, and lamp in correct positions for trial 1.

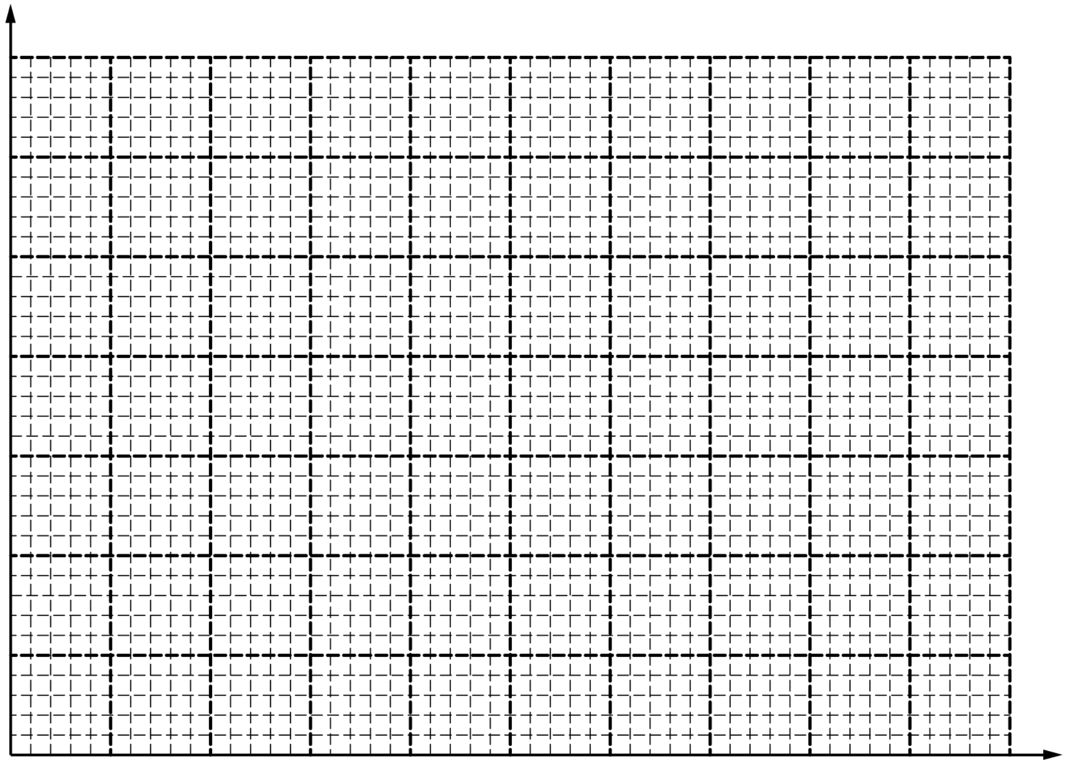


(b) What quantities could be plotted to create a linear graph that could be used to calculate the focal length of the lens? Justify your answer.

Use the empty rows of the table above to record any values you need to calculate.

(c)

i. On the axes below, plot data for the quantities you indicated in part (b). Label and scale the axes.



ii. Use your graph to calculate the focal length of the lens. Show your calculations.

Question 3 continues on the next page.

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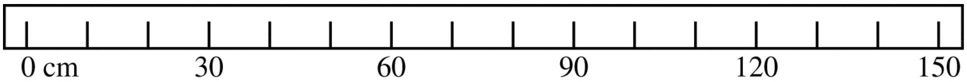
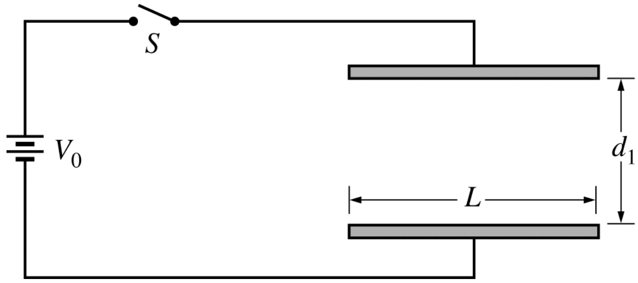
(d) The students are now given a converging mirror of focal length 15 cm to replace the lens and asked to make measurements that would allow them to verify the focal length of the mirror. Describe a procedure that would result in creation of a real image from which the focal length could be determined. On the figure below, representing the optical bench, draw and label the screen, lamp, and mirror in appropriate positions. Indicate what measurements should be made and how they could be used to calculate the focal length.



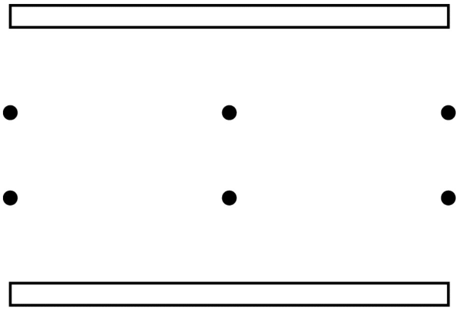


4. (10 points, suggested time 20 minutes)

An uncharged air-filled capacitor has square plates of side L whose separation can be varied. The initial plate separation is d_1, which is comparable to L. The capacitor is connected to a battery with potential difference V_0 and a switch S, as shown above. Assume the dielectric constant of the air is 1.0.

(a) The switch is closed and remains closed for a long time.

i. On the diagram below, draw a vector at each dot to represent the electric field created by the charge on the capacitor plates.



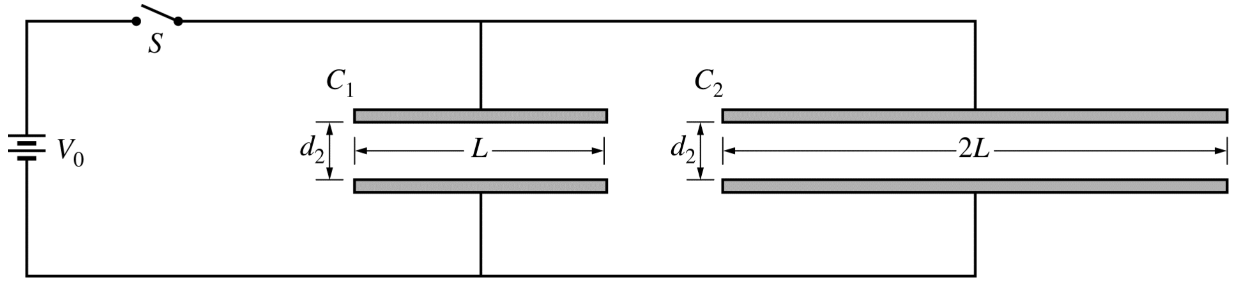
ii. The plates are now moved to a new separation $d_2 << d_1$ and allowed to reach equilibrium. Calculate the magnitude of the charge on each plate of the capacitor in terms of V_0, L, d_2, and physical constants, as appropriate.

(b) The plates are pulled apart to a separation of $2d_2$, where $2d_2 << d_1$.

i. Suppose the switch remains closed while the plates are pulled apart. Compare the magnitude of the new electric field between the plates when their separation is $2d_2$ to the magnitude of the field for plate separation d_2. Support your comparison using appropriate physics principles.

ii. Suppose the switch is opened before the plates are pulled apart. Compare the magnitude of the new electric field between the plates when their separation is $2d_2$ to the magnitude of the field for plate separation d_2. Support your claim using appropriate physics principles.

(c) The new circuit shown below is now assembled. The plates of the original capacitor, C_1, are at separation d_2. A second capacitor, C_2, with square plates of side $2L$ and plate separation d_2, is connected in parallel with the original capacitor. Both capacitors are initially uncharged. The switch is closed and the circuit reaches equilibrium. Let Q_0 be the magnitude of the charge on each plate of capacitor C_1 in the previous circuit, as determined in part (a)(ii). In terms of Q_0, calculate the magnitude of the charge on each plate of the two capacitors in the new circuit.



Answer Key for AP Physics 2 Practice Exam, Section I

Question 1: D
Question 2: B
Question 3: C
Question 4: D
Question 5: B
Question 6: C
Question 7: B
Question 8: D
Question 9: D
Question 10: A
Question 11: C
Question 12: C
Question 13: D
Question 14: C
Question 15: D
Question 16: A
Question 17: C
Question 18: C
Question 19: C
Question 20: D
Question 21: C
Question 22: B
Question 23: D
Question 24: A
Question 25: D
Question 26: C
Question 27: D
Question 28: D
Question 29: D
Question 30: B
Question 31: D
Question 32: A
Question 33: D
Question 34: B
Question 35: C
Question 36: A
Question 131: B, C
Question 132: A, D
Question 133: A, C
Question 134: A, C

2017 AP Physics 2: Algebra-Based
Question Descriptors and Performance Data

Multiple-Choice Questions

Question	Learning Objectives	Essential Knowledge	Science Practices	Key	% Correct
1	2-7.C.1.1	7.C.1	1.4	D	94
2	2-6.A.1.2	6.A.1	1.2	B	84
3	2-3.C.2.2|2-3.G.1.2	3.C.2|3.G.1	7.2|7.1	C	82
4	2-1.A.4.1|2-5.B.8.1|2-7.C.4.1	1.A.4|5.B.8|7.C.4	1.1|1.2|1.1|1.2	D	65
5	2-2.D.3.1	2.D.3	1.2	B	75
6	2-4.E.3.4	4.E.3	1.1|1.4|6.4	C	75
7	2-2.C.5.3	2.C.5	1.1|7.1	B	74
8	2-5.B.9.5|2-5.B.9.6	5.B.9	6.4|2.2	D	70
9	2-3.B.1.4|2-3.C.3.1|2-4.E.2.1	3.B.1|3.C.3|4.E.2	6.4|7.2|1.4|6.4	D	18
10	2-5.B.7.3	5.B.7	1.4|2.2	A	60
11	2-5.B.6.1	5.B.6	1.2	C	52
12	2-5.B.7.1|2-5.B.7.3	5.B.7	6.4|1.4|2.2	C	58
13	2-6.C.1.2|2-6.C.1.1	6.C.1	6.4|1.4	D	69
14	2-4.E.5.3|2-5.B.9.4|2-5.B.9.7	4.E.5|5.B.9	5.1|5.1|5.1	C	38
15	2-4.E.5.2|2-4.E.5.3	4.E.5	6.4|5.1	D	39
16	2-5.B.4.1|2-7.A.2.1	5.B.4|7.A.2	6.4|7.2|7.1	A	55
17	2-5.D.2.6	5.D.2	6.4|7.2	C	45
18	2-4.C.4.1	4.C.4	2.2|7.2	C	58
19	2-7.C.3.1	7.C.3	6.4	C	41
20	2-6.E.5.1	6.E.5	1.4|2.2	D	65
21	2-2.C.2.1|2-2.E.2.1	2.C.2|2.E.2	2.2|6.4|6.4	C	41
22	2-2.C.1.2|2-2.E.3.1|2-2.E.3.2|2-5.B.5.5	2.C.1|2.E.3|5.B.5	2.2|2.2|1.4|6.4|2.2|6.4	B	56
23	2-3.C.2.3	3.C.2	2.2	D	56
24	2-7.A.3.1|2-7.A.3.3	7.A.3	6.4|7.2|5.1	A	59
25	2-4.E.3.3	4.E.3	1.1	D	49
26	2-1.E.3.1	1.E.3	5.1	C	49
27	2-7.A.1.2	7.A.1	1.4|2.2	D	52
28	2-5.F.1.1	5.F.1	2.2	D	71
29	2-3.C.4.2	3.C.4	6.2	D	65
30	2-6.F.1.1|2-6.F.2.1	6.F.1|6.F.2	6.4|7.2|1.1	B	49
31	2-4.E.5.1	4.E.5	2.2|6.4	D	40
32	2-7.A.2.2	7.A.2	7.1	A	54
33	2-3.C.3.1	3.C.3	1.4	D	47
34	2-2.C.1.1	2.C.1	6.4	B	37
35	2-5.B.9.5	5.B.9	6.4	C	36
36	2-6.G.1.1	6.G.1	6.4|7.1	A	31

2017 AP Physics 2: Algebra-Based
Question Descriptors and Performance Data

Question	Learning Objectives	Essential Knowledge	Science Practices	Key	% Correct
131	2-7.A.3.2	7.A.3	4.2	B,C	88
132	2-6.E.2.1|2-6.E.4.1	6.E.2|6.E.4	6.4|4.1|5.1	A,D	70
133	2-6.E.3.3	6.E.3	6.4	A,C	54
134	2-5.B.10.4	5.B.10	6.2	A,C	50

Free-Response Questions

Question	Learning Objectives	Essential Knowledge	Science Practices	Mean Score
1	2-1.E.1.2|2-3.B.1.4|2-3.B.2.1|2-3.C.4.1|2-5.B.10.1|2-5.B.10.2	1.E.1|3.B.1|3.B.2|3.C.4|5.B.10	6.4|6.4|1.1|2.2|6.1|2.2|2.2	4.05
2	2-4.C.3.1|2-5.B.4.1|2-5.B.5.6|2-5.B.6.1|2-5.B.7.1|2-5.B.7.2|2-5.B.7.3|2-7.A.2.2|2-7.A.3.3	4.C.3|5.B.4|5.B.5|5.B.6|5.B.7|7.A.2|7.A.3	6.4|6.4|7.2|5.1|1.2|6.4|1.1|1.4|2.2|7.1|5.1	6.86
3	2-6.E.4.1|2-6.E.5.2	6.E.4|6.E.5	4.1|5.2|4.1|5.1	4.86
4	2-2.C.5.1|2-2.C.5.2|2-4.E.4.1|2-4.E.5.1|2-4.E.5.2	2.C.5|2.C.5|4.E.4|4.E.5	1.1|2.2|2.2|6.4|6.4|6.4	2.43

Question 1

10 points total | **Distribution of points**

(a) 5 points

For indicating that a floating object displaces its weight in water	1 point
For indicating that a submerged object displaces its volume in water	1 point
For justifying with correct reasoning that the volume of displaced water required to make the cube float is much greater than the cube's volume Examples: Noting that the density of the block is much greater than that of water; noting that when the cube is on the bottom the buoyant force is less than the weight because of the normal force	1 point
For predicting that the water level is lower when the block is submerged (or higher when the block is floating) given some attempt to explain the concepts listed above	1 point
For a logical, relevant, and internally consistent response that addresses the required argument or question asked, and follows the guidelines described in the published requirements for the paragraph-length response	1 point

(b)

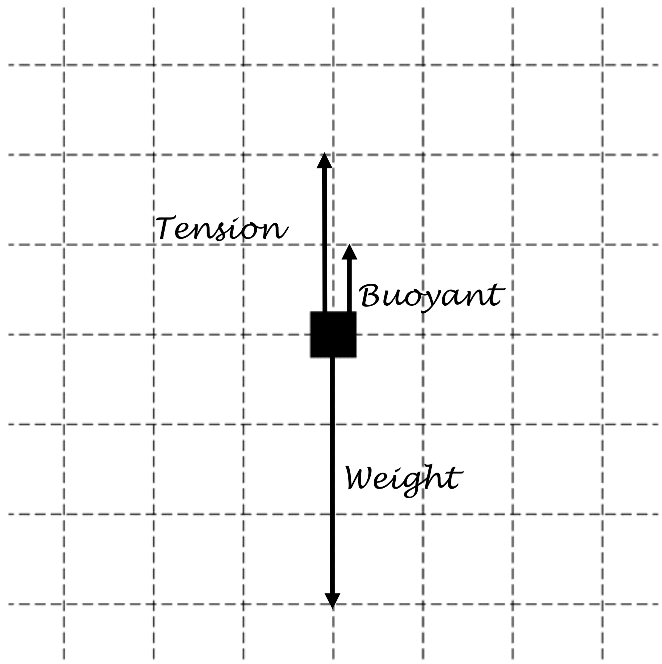
i) 1 point

For indicating that the tension is the same at any depth with a correct reference to the buoyant force Example: The volume of water displaced remains the same so the buoyant force remains the same. The weight of the cube does not change, so the force exerted by the crane is unchanged.	1 point

Question 1 (continued)

Distribution of points

ii) 2 points



For a labeled buoyant force directed upward and a labeled gravitational force directed downward, with the buoyant force drawn shorter than the gravitational force	1 point
For a labeled tension force directed upward and the sum of the drawn vectors approximately indicating equilibrium	1 point

Figure can have a single buoyant force, or a set of pressure forces that reasonably equals an appropriate net buoyant force, but not both.

Only force vectors that clearly originate on the block earn credit.

iii) 2 points

For an expression obtained by using Newton's 2nd law for equilibrium and the free body diagram in part (b)(ii)	1 point

$F_{Tension} = F_g - F_b$

$F_{Tension} = m_{cube}g - \rho g V_{cube}$

For correct substitutions into the buoyant force and the gravitational force expressions or a numerical answer implying correct substitutions of the given quantities	1 point

$F_{Tension} = (730\ \text{kg})\left(9.8\ \text{m}/\text{s}^2\right) - \left(1000\ \text{kg}/\text{m}^3\right)\left(9.8\ \text{m}/\text{s}^2\right)(0.45\ \text{m})^3$

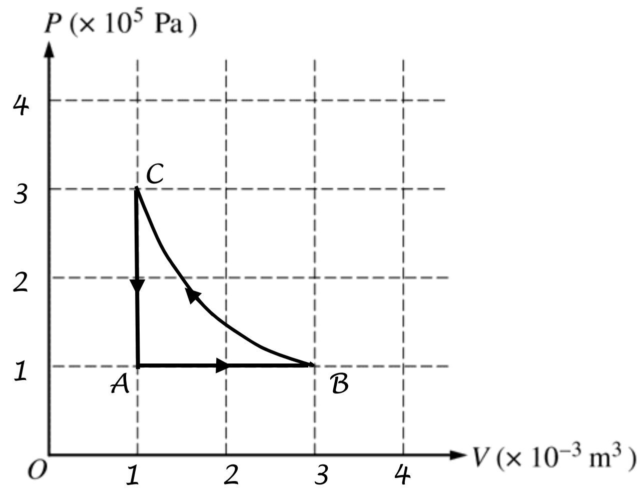
$F_{Tension} = 6300\ \text{N}$ (or 6400 N using $g = 10\ \text{m}/\text{s}^2$)

Question 2

12 points total **Distribution of points**

(a)

i) 3 points



For a straight line at constant pressure from A to B to the correct final volume, which must be labeled 3 — 1 point

For an isotherm from B to C ending at the right final pressure, which must be labeled 3; curve must be obviously concave up — 1 point

For a straight line from C to A at constant volume, with state A labeled (1, 1) — 1 point

ii) 1 point

Using the ideal gas law:

$P_A V_A = nRT_0$

$P_B V_B = P_A 3V_A$

$nRT_B = 3nRT_0$

For the correct answer — 1 point

$T_B = 3T_0$

iii) 1 point

For a correct comparison of the temperatures or correctly indicating — 1 point

$T_B = T_C = 3T_A$ or $3T_0$ or a value consistent with the answer to part (a)(i) or (ii)

AND making a reference to the isotherm or the values of P and V at points C and A

Examples:

$T_C = 3T_0$ since it is on an isotherm with point B, as shown by the graph.

On the graph, C has the same volume and 3 times the pressure as A. By the ideal gas law it thus has a temperature 3 times that of A.

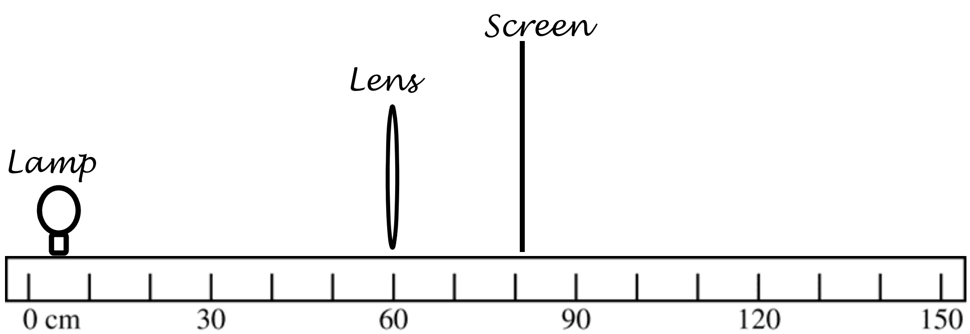
Question 2 (continued)

	Distribution of points
(b)	
i) 2 points	
For recognizing that the volume remains constant and describing a valid way to accomplish this in the procedure	1 point
For recognizing that the temperature must decrease and describing a valid way to accomplish this in the procedure	1 point
Example: Lock the piston in place and put the cylinder in a cool water bath.	
ii) 2 points	
For some reasonable description of microscopic processes in which collision of molecules is the process by which energy is transferred	1 point
For correctly stating the direction of energy flow (e.g. hot (fast) to cold (slow)) and linking the direction of energy flow to the final temperature of the gas	1 point
Example: The warmer gas (faster molecules) collide with the cooler surroundings (slower molecules). The net energy flow is from the gas to the surroundings which results in cooler gas (slower molecules).	
(c)	
i) 2 points	
For numerically calculating the amount of work done $(W = \lvert P\Delta V \rvert)$	1 point
$W = -P\Delta V = -\left(1 \times 10^5 \text{ Pa}\right)\left(2 \times 10^{-3} \text{ m}^3\right)$	
For the correct answer with negative sign (no units required)	1 point
$W = -200 \text{ J}$	
ii) 1 point	
For applying conservation of energy using the value from part (c)(i)	1 point
$\Delta U = Q + W$	
$300 \text{ J} = Q - 200 \text{ J}$	
$Q = 500 \text{ J}$	

Question 3

12 points total **Distribution of points**

(a) 2 points



For the lamp correctly positioned 55 cm from the converging lens — 1 point

For the screen correctly positioned 21.3 cm from the lens, on the opposite side from the lamp — 1 point

(b) 3 points

For indicating a valid pair of quantities that would lead to a linear graph from which the focal length could be determined — 1 point

For justifying the choice of quantities, either by indicating how to use the slope or intercept of the line to determine the focal length or by indicating how the lens equation yields a linear function — 1 point

For calculating data for the table, with at least three correct pairs of numbers — 1 point

Table below shows data for two possibilities: $1/s_0$ and $1/s_i$; $s_0 s_i$ and $s_0 + s_i$

Trial	1	2	3	4	5
Object Distance s_o (cm)	55	45	35	30	25
Image Distance s_i (cm)	21.3	23.8	28.5	32.5	43
$1/s_o$	0.018	0.022	0.029	0.033	0.04
$1/s_i$	0.047	0.042	0.035	0.031	0.023
$s_o s_i$	1172	1071	998	975	1075
$s_o + s_i$	76	69	64	63	68

(c)

i) 2 points

For the axes scaled linearly and labeled with units for plotted quantities (even if the wrong quantities are plotted), such that the plot uses at least half of the axes — 1 point

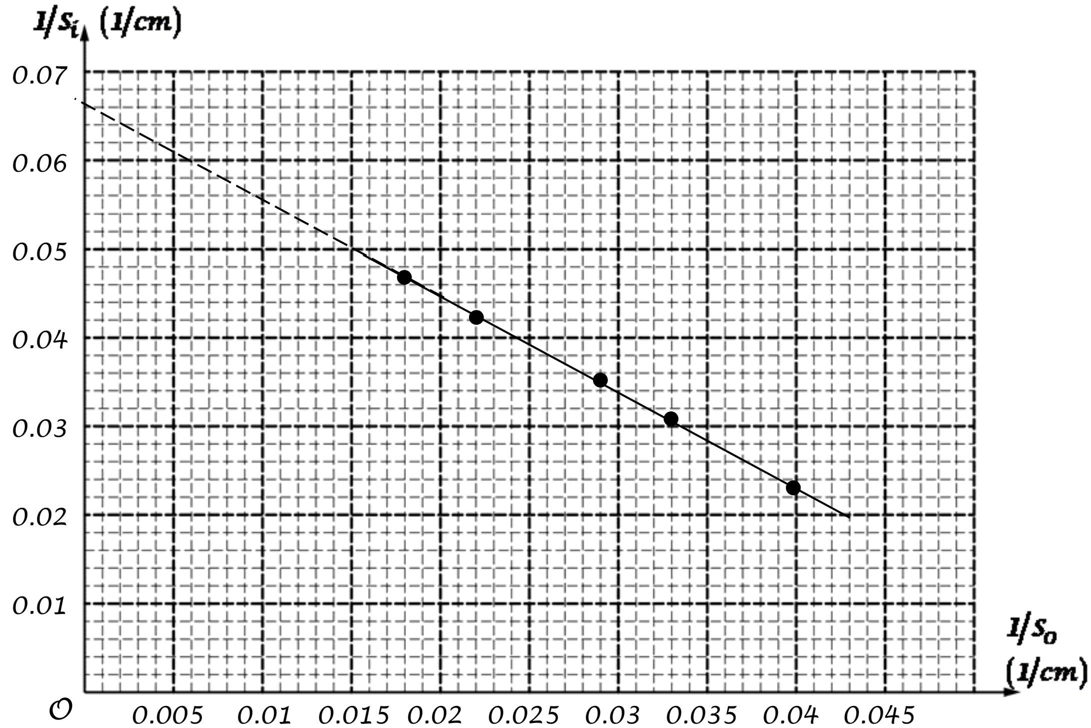
For correctly plotting the data listed in the part (b) table — 1 point

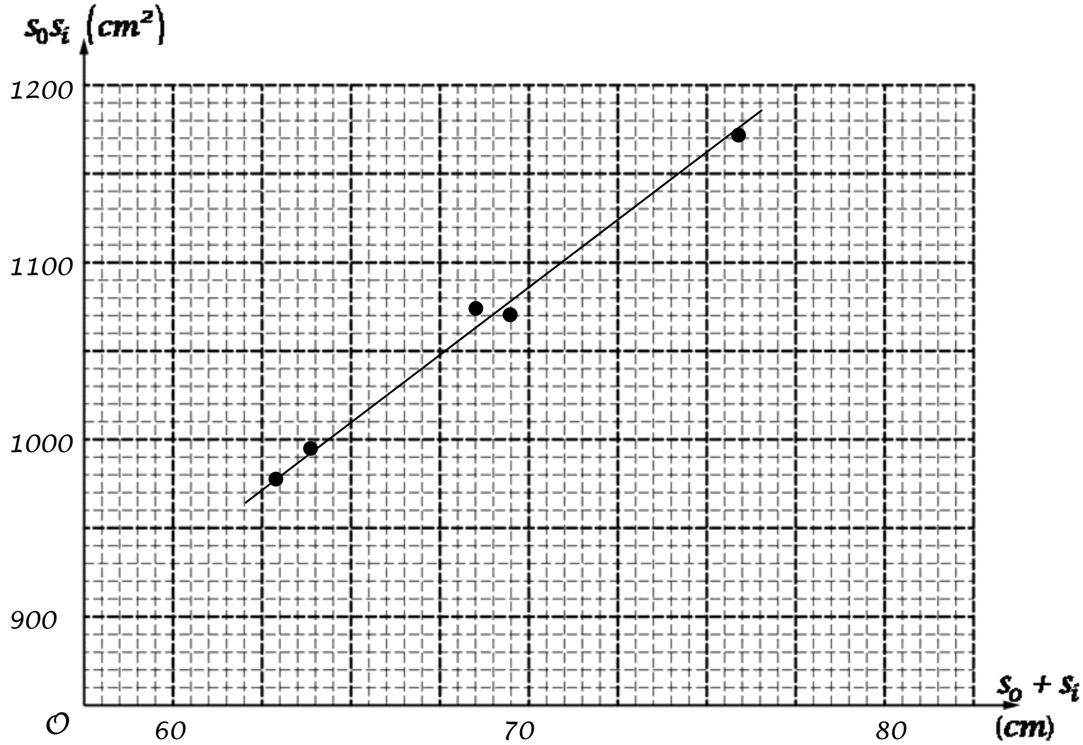
The following graphs show the two possibilities noted above.

Question 3 (continued)

Distribution of points

(c) i) (continued)





Question 3 (continued)

Distribution of points

(c) (continued)

ii) 2 points

For drawing a straight best-fit line accurately through the data (data points evenly distributed about the line and representing the trend of the data) — 1 point

For accurately calculating the slope or intercept using the graph (depending on quantities plotted) and relating it to the focal length — 1 point

For the first graph ($1/s_\mathrm{i}$ as a function of $1/s_0$):

From the lens equation $\frac{1}{s_i} = \frac{1}{f} - \frac{1}{s_0}$, so $\frac{1}{f}$ equals the intercept on the $1/s_\mathrm{i}$ axis

$$\frac{1}{f} = 0.066\ \mathrm{cm}^{-1}$$

$$f = 15.2\ \mathrm{cm}$$

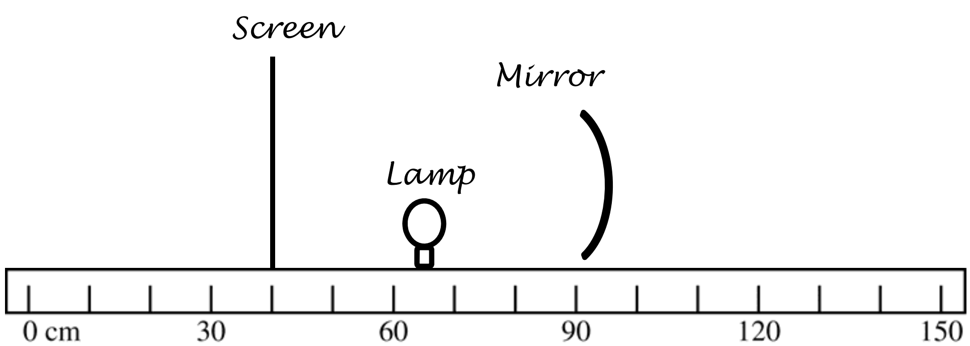
For the second graph ($s_0 s_i$ as a function of $s_0 + s_i$):

The lens equation can be rearranged as $s_0 s_i = f\left(s_0 + s_i\right)$, so f equals the slope

$$f = \frac{(1170 - 970)\ \mathrm{cm}^2}{(75.5 - 62.5)\ \mathrm{cm}} = \frac{200}{13}\ \mathrm{cm}$$

$$f = 15.4\ \mathrm{cm}$$

(d) 3 points



For drawing and labeling the lamp, screen, and converging mirror at positions that will create a real image — 1 point

For describing appropriate measurements — 1 point

For describing a calculation of the focal length that correctly uses the measurements — 1 point

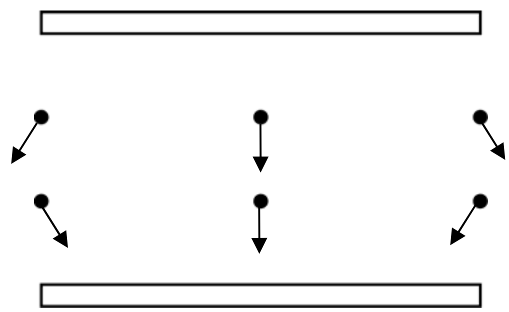
Example: Place the lamp between 15 and 30 cm from the mirror. Place the screen on the other side of the lamp from the mirror, adjusting its position until the clearest image is formed. Measure the distance of the lamp and screen from the mirror. Use the equation $\frac{1}{s_i} + \frac{1}{s_0} = \frac{1}{f}$ to calculate the focal length.

Question 4

10 points total | **Distribution of points**

(a)

i) 2 points



For a representation of the electric field using vectors (not curved lines) that each have a component downward — 1 point

For accurately representing fringing near the edges and a uniform field near the middle (even if field lines are drawn instead of vectors, or the field has components upward instead of downward) — 1 point

ii) 1 point

For a correct calculation of the charge — 1 point

$Q = C\,\Delta V = \left(\varepsilon_0 L^2 / d_2\right) V_0$

$Q = \varepsilon_0 L^2 V_0 / d_2$

(b)

i) 2 points

For indicating that since the battery remains connected, the potential difference across the capacitor remains the same — 1 point

For indicating that the electric field must decrease since $E = \Delta V/d$ and d has increased — 1 point

Alternate Solution — *Alternate points*

$C = \kappa\varepsilon_0 A/d$

For reasoning that capacitance decreases as d increases — *1 point*

$Q = C\,\Delta V$

For reasoning that the potential difference across the capacitor remains the same and therefore the charge decreases — *1 point*

$E = Q/\varepsilon_0 A$ *, so E decreases because Q decreases*

Question 4 (continued)

	Distribution of points
(b) (continued)	
ii) 2 points	
For indicating that the charge on each plate is the same as in part (a)(ii)	1 point
For reasoning that the electric field depends only on the magnitude of the charge, and therefore does not change (or has a change consistent with the change in charge indicated)	1 point
Examples:	
The charge on the plates does not change, so E remains the same.	
$E = \Delta V/d = Q/Cd$. Since $C \propto 1/d$, E is independent of the plate spacing which is the only thing that changes.	
Alternate Solution	*Alternate points*
$C = \kappa\varepsilon_0 A/d$ *, so as d increases C decreases*	
$\Delta V = Q/C$	
For recognizing that since charge is constant, as C decreases ΔV *increases*	*1 point*
$E = \Delta V/d$	
For reasoning that E does not change since ΔV *and d increase by the same factor*	*1 point*
(c) 3 points	
For indicating that the potential difference is the same across each capacitor	1 point
For indicating that $C_2 = 4C_1$	1 point
For values of the charge on each capacitor consistent with the relationship between capacitances	1 point
$Q_1 = Q_0$, $Q_2 = 4Q_0$	
Example: The conditions for C_1 are the same as in part (a)(ii), so it has a charge Q_0. The area of C_2 is four times that of C_1, so it has four times the capacitance. In order to have the same potential difference as C_1, it must therefore have four times the charge.	
Alternate solution	*Alternate Points*
For indicating the potential difference is the same across each capacitor	*1 point*
For indicating $A_2 = 4A_1$	*1 point*
For values of the charge on each capacitor consistent with the relationship between areas	*1 point*
$Q_1 = \varepsilon_0 V_0 L^2/d_2$	
$Q_2 = \varepsilon_0 V_0 (2L)^2/d_2$	